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SUMMARY OF QUALIFICATIONS

Motivated graduate of Computer Science transitioning from chemical industry work to software engineering and development jobs with 3-4 years of experience, both in academic and industry combined. Developing new technology solutions in my current Senior Lab Technician position to reduce data entry errors with C# .NET and PowerShell applications. Improving lab technician tasks of sample analysis such as reducing processing time and adapting new technology techniques for driving delivery of laboratory clients' requests. Consistently using JavaScript, Python, C# .NET with the software development lifecycle for full-stack development of project applications.

EDUCATION

Lamar University, Beaumont TX — May 2024
Bachelor of Science in Computer Science — GPA:3.23

The University of Texas at San Antonio, San Antonio TX — December 2017
Bachelor of Science in Biology, minor in Chemistry — GPA:3.17

TECHNICAL SKILLS

Programming Languages: Java, C++, C#, C, Python, JavaScript, HTML, PowerShell
Libraries and Frameworks: TypeScript, Matplotlib, NumPy, Pandas, Node.JS, React.JS, CSS, Svelte, .NET
Tools and Platforms: Docker, Tableau, Git, Visual Studio, Visual Studio Code, Ghidra
Databases and Query: MySQL, SQLite3
Cloud Services: AWS Lambda, AWS Translate, AWS Transform, AWS Comprehend, SAM
Linux OS: Kali Linux, Fedora

RELEVANT EXPERIENCE

Senior Lab Technician | Southwest Research Institute — May 2019- Present

- Demonstrated ASTM gas chromatography techniques and methods for sample analyses adhering to best lab practices for clients.
- Microsoft Office products were used to digitize documentation, optimizing lab technician workflow and improving turnaround time and sample testing utilization rates.
- Improved training comprehension for colleagues on analytical chemistry test protocols, resulting in increased client satisfaction and growing sample analysis throughput.

Software Automation Developer Volunteer, Div 08 | Southwest Research Institute — January 2023 - Present

- Designed, supported, and tested software solutions to automate laboratory workflows, increasing operational efficiency and data accuracy for analytical chemistry testing.
- PowerShell scripting and C# .NET programming to automate and improve technical workflows in ASTM methodology to reduce redundancies and lessen human error during data reporting.
- Collaborated with Lead IT Analyst in improving personal skills for the software development lifecycle and object-oriented design of laboratory programs.
- Utilized version control with Git to maintain and increase efficiency in cross-functional duties, while maintaining documentation of improvements to programs to reduce human error from manual tasks.

PROJECTS

Blood Cell Morphology Analysis Software — ***Python, JavaScript, Jupyter***

- Developed computer vision analysis on blood cell pictures to establish classifications on types of blood cells and amount in an image within a dataset.
- Evaluation of cells determines how to further go next with model limitations, dataset bias, false negatives, and validation.

Convention Event Registration Software ***Docker, Python, Svelte***

- Collaborated with a small team in creating new event registration software, focusing on designing and implementing functional API calls in Python from front-end to back-end communication.
- Communicated with leaders of the project to establish the design of web pages and improve API call functionality.
- Optimized error logging in JavaScript and testing with Postman to reduce load times and improve the dependability of web pages designed in Svelte.

Mandarin Sentence Constructor (exampro.co in Development) | Link ***AWS Services, SQLite3, JavaScript***

- Designed a serverless language translation application with AWS services and SAM model to detect input language and translate into selected languages, improving accuracy and speed of translations with a focus on the Mandarin Chinese language.
- Implemented JavaScript grammar validation algorithms designed to reduce the time taken to translate and present to the user.
- Enhanced the SQLite3 database schema for vocabulary management, supporting scalable and maintainable feature growth.

Weather API Project | Link ***Python***

- Developed a weather and airline application in Python to focus on REST API data organization to efficiently view current weather and airline conditions.
- Modularized system by integrating web development tools such as Flask to enable an increase in user engagement and interface usability.
- Improved API data parsing and error handling to reduce delivered times for data delivery and create more robust accuracy for weather data.

Vehicle Sales Project ***Java, MySQL***

- Designed mock vehicle sales user interface with secure user authentication and role-based access control in Java, strengthening application security and data integrity.
- Optimized MySQL queries for faster search and transaction processing, improving overall system performance.
- Applied object-oriented design principles in Java to ensure best secure software practices and code scalability and maintainability for future development.

Chatroom Socket Programming Project | Link ***C***

- Developed persistent user profile management in C, enabling unique usernames and session continuity across chat sessions.
- Engineered multi-threaded server logic in C to support concurrent client connections for scalability.
- Strengthened socket communication protocols in C to ensure stable and secure data exchange between clients and server.

ADDITIONAL EXPERIENCE

UTSA Hackathon, University of Texas at San Antonio, San Antonio, TX March 2020

Assistant Programmer to Lead Programmer

- Co-developer of Java project modifications to visualize social distancing simulation of COVID-19's March 2020 rate of contraction of COVID-19 in a 3D video game based on age and chance of contraction data produced by the CDC. Implemented Java fundamentals to learn quick implementations of Java Sponge API to improve accurate data representation in the simulation.

PROFESSIONAL ORGANIZATIONS

Member, **Association for Computing Machinery**, Lamar University January 2020 – May 2024